MILITARY SPECIFICATION

BATTERY, HED, SINGLE CELL, PRIMARY RESERVE

This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly, and packing of the HED, single cell, primary reserve battery.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance
MIL-P-63238	-	Projectile, 155MM, AT, M718/M741 Loading, Assembling and Packing of
MIL-T-63420	-	Thionyl Chloride Electolyte
MIL-P-63430	-	Projectile, 155MM, Test, Non-Haz, M718/M741 Types Loading, Assembling and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-331	-	Fuze and Fuze Components Environmental and Performance Tests for
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9317598	-	Projectile, 155MM Test, Non-Haz, M741 Type
9317608	-	Mine, Test, Non-Haz (M741)
9328787	-	Battery HED, Single Cell, Primary Reserve

INSPECTION EQUIPMENT DRAWINGS

9275628	-	Firing Pin
9275634	-	Procedure, Assembly, Power Source
9287150	-	M718 Electronics Assembly
9287151	-	M741 Electronics Assembly
9275635	-	Modified Molding, Power Source
9296892	-	Primer, Stab

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 493 - Standard Methods of Test for Leaks Using the Mass Spectrometer Leak Detector in the Inside-Out Testing Mode.

(Application for copies of ASTM publication should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9328787 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Non-activation voltage. The non-activation voltage of the battery, recorded at:

Relative Humidity (RH)			Millivolt
a. Max	40%	shall be less than	5mV
b. Greater than	45%	shall be at a max of	200mV

All measurements are to be recorded with an approved millivoltmeter having a minimum impedance of 10^7 ohm.

3.4 Battery voltage rise time. The battery voltage shall rise to minimum voltage of 2.5 volts in a maximum of one half (0.5) second after activation over the temperature range of -45°F to +145°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.5 Battery capacity. The battery shall remain above 2.5 volts for a minimum of two hundred and forty (240) hours when subjected to a 6.6K ohm ± 3% resistive load over the temperature range of -45°F to +145°F.

3.6 Leak test. The batteries shall meet the leak rate requirements as specified on the drawing (9328787) when tested in accordance with the provisions of paragraph 4.4.6.5.

3.7 Weld integrity.

3.7.1 Anode lead to terminal pin weld. The anode lead to terminal pin weld, when pulled to destruction, shall not result in separation of the weld. Lead material shall fail before the weld.

3.7.2 Anode and cathode lead to screen welds. The anode and cathode lead to screen weld, when pulled to destruction, shall not result in separation of the weld. Parent material (screen or lead) shall fail before the weld.

3.7.3 Case to terminal plate weld. The case to terminal plate weld shall not show evidence of pits or cracks when visually examined.

3.8 Terminal plate glass to metal seal. The glass seal shall not have cracks on both sides of the pin or any one crack that extends over one-third the distance from the pin to the plate.

3.9 Radiographic examination. X-ray negatives of the batteries shall be examined for the following:

a. There shall be continuity of the anode lead.

b. There shall be no sharp bends in the anode lead.

c. The anode lead shall not be bent against the outside case.

d. The ampul shall be intact and show evidence of electrolyte.

3.10 Heat soak. The ampul assembly, electrolyte (9328782) shall withstand a 24 hour heat soak at 165 ± 5°F without leakage or breakage.

3.11 Environmental tests. The batteries shall meet the requirements of paragraphs 3.3, 3.4, 3.5 before and after being subjected to the following tests:

a. Shock. Drop per MIL-STD-331, Test 111, except batteries shall be mounted in hardwood block of 40 cu. inches weighted to 5 pounds and shall be dropped from the height of 7 ft.

b. Transportation vibration. Per MIL-STD-331, Test 123.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:27Z
Check the source to verify that this is the current version before use.

3.12 Ballistic rail gun test. The batteries shall meet the requirements of paragraph 3.3 prior to and after being subjected to 10,000 + 800/-300g's when tested in accordance with the provisions of paragraph 4.4.6.8.

3.13 Ballistic functioning. The batteries shall function properly when assembled and tested in 155MM Test Non-Haz, M718/M741 Types at Zone 7 charge for the M109A1.

3.14 Ampul assembly electrolyte condition. The electrolyte when examined inside the ampul shall be clear and of a colorless or light yellow to light pink appearance.

3.15 Hardness of case. The case (9275592) shall have a Knoop hardness of 231 max at 200 gm load, or 96 Rockwell B (max.).

3.16 Surveillance. The contractor shall provide for surveillance quantities of batteries from each months production. A total quantity of 160 batteries per lot shall be supplied in accordance with 4.5.7.

3.17 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.18 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. Metal parts shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, corrosion products, and other defects or foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

and tested and found to comply

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the items in sample quantities as indicated.

Part Description	Drawings	Quantity
Battery, HED, Single Cell Primary Reserve	9328787	438

4.3.2 Inspections to be performed. See MIL-A-48078, and paragraphs 4.4.2, testing per paragraph 4.4.3, and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.4	Battery, HED, Single Cell Primary Reserve		9328787
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Battery, Single Cell, Primary Reserve				
	Examination of defects	438		3.2	4.4.2.1
	X-ray examination	126		3.9 Note 1	4.4.6.3
	Leak test	126		3.2,3.6	4.4.6.5
	Non-activation voltage	126		3.3	4.4.6.1
	Voltage rise time and capacity test	72		3.4,3.5	4.4.6.2.1
	Electrical short, anode to cathode	126		3.2	4.4.6.12
	Environmental Test	24		3.11	4.4.6.7
	7-ft. drop	12		3.11.a	4.4.6.7.a
	Transportation vibration	12		3.11.b	4.4.6.7.b
	Ballistic Rail Gun Test	72		3.12	4.4.6.8
	Ballistic Test	216		3.13	4.4.6.9

NOTES: (1) Information only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of batteries shall contain thionyl chloride electrolyte (MIL-T-63420) of one lot interfix from one manufacturer.

Maximum lot size shall be no more than one (1) months production (30 calendar day period).

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Battery HED, Single Cell Primary Reserve				9328787
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak test		100%	3.2,3.6	4.4.6.5
102	Electrical short anode to cathode		100%	3.2	4.4.6.12
103	Shim, ampul support missing		100%	3.2	Visual
104	Disc, insulator missing		100%	3.2	Visual
105	Pad, ampul support missing		100%	3.2	Visual
106	Glass to metal seal		0.40%	3.8	Visual
107	Lead to terminal plate weld checks (prior to terminal plate closure)	3	--	3.7	4.4.3.2
108	7 Foot Drop	12	--	3.11	4.4.6.7.a
109	Radiographic examination	72	--	3.9, Note 1	4.4.6.3
110	Non-activation voltage	See	See	3.3	4.4.6.1
111	Voltage rise time	Para	Para	3.4	4.4.6.2.2
112	Capacity	4.4 3.1	4.4 3.1	3.5	4.4.6.2.2
Minor					
201	Length incorrect		0.65%	3.2	Gage
202	Diameter incorrect		0.65%	3.2	Gage
203	Case to terminal plate weld improper		0.65%	3.7.3	Visual (2X)
204	Poor workmanship		0.65%	3.18	Visual

NOTES: (1) Radiographic examination shall be for information only.

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Terminal Plate Assembly - Machined	1 OF 1	9328777
			NEXT HIGHER ASSEMBLY 9328787

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter incorrect		0.40%	3.2	Gage
Minor					
201	True position of OD with terminal pin		0.65%	3.2	Gage
202	Surface finish improper		0.65%	3.2	Visual
203	Height of terminal pin above inside shoulder incorrect		0.65%	3.2	Gage
204	Thickness of terminal plate incorrect		0.65%	3.2	Gage
205	Height of terminal pin above outside shoulder incorrect		0.65%	3.2	Gage
206	Pin surface flat		0.65%	3.2	Visual
207	Taper not present		0.65%	3.2	Visual
208	Glass extending above any metal surface		0.65%	3.2	Visual
209	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Pin, Terminal	1 OF 1	9328770
			NEXT HIGHER ASSEMBLY: 9328777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Diameter incorrect		0.65%	3.2	Gage
202	Surface finish improper		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9328780
4.4.2.4	Cell Assembly				NEXT HIGHER ASSEMBLY 9328789
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Anode lead not positioned on inner diameter of anode assembly		0.40%	3.2	Visual
102	Bottom separator missing, damaged, or not completely covering bottom insulator (Insulator visible)		0.40%	3.2	Visual
103	Anode assembly not uniformly expanded against the vertical separator		0.40%	3.2	Visual
Minor					
201	Opening of anode assembly aligned with opening of cathode		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Anode Assembly		9328779
			NEXT HIGHER ASSEMBLY
			9328780

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any evidence of the screen exposed at the edges		0.40%	3.2	Visual
102	Lead side of screen pressed onto anode pad		0.40%	3.2	Visual
Minor					
201	Length incorrect		0.65%	3.2	Gage
202	Width incorrect		0.65%	3.2	Gage
203	Thickness incorrect		0.65%	3.2	Gage
204	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Screen Assembly, Anode	1 OF 1	9275574
			NEXT HIGHER ASSEMBLY 9328779

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of lead		0.40%	3.2	Gage
102	Length of lead from screen incorrect		0.40%	3.2	Gage
103	Length of lead overlap on screen incorrect		0.40%	3.2	Gage
104	Weld missing or improper		0.40%	3.2	Visual
105	Weld integrity	See 4.4.3.2.2	--	3.7.2	Test
Minor					
201	Screen length incorrect	See 4.4.4	0.65%	3.2	Gage
202	Screen width incorrect		0.65%	3.2	Gage
203	Lead thickness incorrect		0.65%	3.2	Gage
204	Lead width incorrect		0.65%	3.2	Gage
205	Lead material improper			3.2	Certification of Conformance
206	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Disc, Insulator	1 OF 1	9328764
			NEXT HIGHER ASSEMBLY 9328787

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Outside diameter incorrect (max)		0.65%	3.2	Gage
202	Inside diameter incorrect (min)		0.65%	3.2	Gage
203	True position of ID with OD		0.65%	3.2	Gage
204	Outside radius incorrect		0.65%	3.2	Gage
205	Thickness incorrect		0.65%	3.2	Gage
206	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Shim, Ampul Support	1 OF 1	9328765
			NEXT HIGHER ASSEMBLY: 9328787

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burrs present		0.40%	3.2	Visual
102	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Outside diameter incorrect (max)		0.65%	3.2	Gage
202	Inside diameter incorrect (min)		0.65%	3.2	Gage
203	Height incorrect (min)		0.65%	3.2	Gage
204	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Ampul Assembly, Electrolyte				9328782
					NEXT HIGHER ASSEMBLY 9328789
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Heat soak		100%	3.10	4.4.6.6
102	Condition of ampul assembly		100%	3.14	Visual
103	Overall length incorrect (max)		0.40%	3.2	Gage
104	Outside diameter incorrect (max)		0.40%	3.2	Gage
105	True position of tip to outside diameter incorrect		0.40%	3.2	Gage
106	Electrolyte volume incorrect		--	3.2	4.4.6.4
Minor					
201	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Separator Assembly		9328793
			NEXT HIGHER ASSEMBLY 9328780

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Vertical separator missing, damaged, or not overlapped at seam		0.40%	3.2	Visual
102	Bottom separator missing or damaged or not completely covering bottom insulator		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.11	Cathode Assembly		12556494
			NEXT HIGHER ASSEMBLY 9328791

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Screen exposed at the edges		0.40%	3.2	Visual
102	Sharp, jagged edges present		0.40%	3.2	Visual
103	Cathode contamination		0.40%	3.2	Visual
104	Blown off area contaminated		0.40%	3.2	Visual
105	Surface dimension of cathode material incorrect		0.40%	3.2	Gage
106	Cathode material and composition improper		0.40%	3.2	Certification of Conformance
107	Cathode weight incorrect		0.40%	3.2	Gage
108	Cathode thickness incorrect		0.40%	3.2	Gage
109	Location of lead incorrect		0.40%	3.2	Gage
110	Weld missing or improper		0.40%	3.2	Visual
111	Weld integrity		--	3.7.2	Test
Minor					
201	Cathode length incorrect		0.65%	3.2	Gage
202	Cathode width incorrect		0.65%	3.2	Gage
203	Cathode assembly thickness incorrect		0.65%	3.2	Gage
204	Length of lead from screen incorrect		0.65%	3.2	Gage
205	Length of lead overlap on screen incorrect		0.65%	3.2	Gage
206	Lead thickness incorrect		0.65%	3.2	Gage
207	Lead width incorrect		0.65%	3.2	Gage
208	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Insulator, Bottom		9328763
			NEXT HIGHER ASSEMBLY 9328791

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Diameter incorrect	See Para. 4.4.4	0.65%	3.2	Gage
202	Thickness incorrect		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Separator, Bottom	1 OF 1	9328762
			NEXT HIGHER ASSEMBLY: 9328793

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Diameter incorrect	See Para. 4.4.4	0.65%	3.2	Gage
202	Thickness incorrect		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.14	Separator, Vertical	1	1	9328761
				NEXT HIGHER ASSEMBLY 9328793

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Grain direction improper		0.40%	3.2	Visual
102	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Length incorrect	See	0.65%	3.2	Gage
202	Width incorrect	Para.	0.65%	3.2	Gage
203	Thickness incorrect	4.4.4	0.65%	3.2	Gage
204	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Case	1 OF 1	9275592
			NEXT HIGHER ASSEMBLY 9328791

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter incorrect		0.40%	3.2	Gage
102	Length incorrect		0.40%	3.2	Gage
103	Thickness incorrect		0.40%	3.2	Gage
104	Discoloration		0.40%	3.2	Visual
105	Sensitizing and/or grain coarsening present		0.40%	3.2	Visual
106	Hardness improper	3	--	3.15	4.4.6.10
107	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Finish improper		0.65%	3.2	Visual/Standard
202	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Barrier, Ampul				9328759
					NEXT HIGHER ASSEMBLY 9328789
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Thickness incorrect (max)	See	0.65%	3.2	Gage
202	Profile incorrect (four places)	Para.	0.65%	3.2	Gage
203	Length incorrect (four places)	4.4.4	0.65%	3.2	Gage
204	Poor workmanship		0.65%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Cell and Ampul Assembly	1 OF 1	9328789
			NEXT HIGHER ASSEMBLY: 9328787

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ampul barrier missing		100%	3.2	Visual
102	Ampul assembly missing or improperly positioned		100%	3.2	Visual
103	Anode lead not positioned on inner diameter of assembly		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Ampul Blank	1 OF 1	9328781
			NEXT HIGHER ASSEMBLY: 9328782

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ampul diameter incorrect		0.40%	3.2	Gage
102	Ampul length to shoulder incorrect		0.40%	3.2	Gage
103	Stricture OD incorrect (2 places)		0.40%	3.2	Gage
104	Stricture ID incorrect		0.40%	3.2	Gage
105	Stricture true position incorrect		0.40%	3.2	Gage
106	Wall thickness incorrect		0.40%	3.2	Gage
107	Bottom thickness incorrect		0.40%	3.2	Gage
108	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Overall length incorrect		0.65%	3.2	Gage
202	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Pad, Ampul Support		9328768
			NEXT HIGHER ASSEMBLY 9328787

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Material improper		--	3.1	Certificate of Conformance
Minor 201	Inside diameter incorrect		0.65%	3.2	Gage
202	Outside diameter incorrect		0.65%	3.2	Gage
203	Thickness incorrect		0.65%	3.2	Gage
204	Poor workmanship		0.65%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Screen, Expanded Metal		9275582
			NEXT HIGHER ASSEMBLY 9275574

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certificate of Conformance
Minor					
201	Thickness incorrect		0.65%	3.2	Gage
202	Mesh dimensions incorrect		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Cathode Case Assembly	1 OF 1	9328791
			NEXT HIGHER ASSEMBLY: 9328793

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of cathode damage		0.40%	3.2	Visual
102	Bottom insulator missing, damaged, or not flat on bottom of case		0.40%	3.2	Visual
103	Cathode grid not against case		0.40%	3.2	Visual
104	Cathode lead missing or improperly positioned		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.18	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63524A(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.22	Lead	1	1	9275575
				NEXT HIGHER ASSEMBLY: 9275574/9328779

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper		--	3.1	Certification of Conformance
Minor					
201	Thickness incorrect		0.65%	3.2	Gage
202	Width incorrect		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Non-activation voltage, battery voltage rise time, battery capacity. Samples of batteries shall be tested for non-activated voltage, rise time and capacity in accordance with Table II.

Performance Samples
TABLE II

Lot Qty	Sample Qty	Non-Activation Voltage (para 3.3, Test at RH, a. & b.)	Rise Time (para 3.4) and Capacitance (para. 3.5)
120 to 1200	72	72-0-1	72-1-2
1201 to 10,000	180	180-1-2	180-2-3
10,001 to 35,000	300	300-1-2	300-3-4

Samples shall be equally divided into three (3) groups: one group tested at -45°F, one group tested at room ambient and one group tested at +145°F. Allowable defects shall be as indicated. Testing for the non-activation voltage shall be performed as specified in 4.4.6.1. Testing for the voltage rise time and voltage capacity shall be performed as specified in 4.4.6.2.2.

4.4.3.2 Weld integrity.

4.4.3.2.1 Lead to terminal plate assembly. At the start of each day and after welding parameters have been set, weld three (3) dummy terminal plates and leads. Pull lead to destruction to determine weld integrity (para. 3.7.1). Do not proceed with production welding until all three (3) dummy units are accepted. At the end of each four (4) hour period during the continuous production run and at the end of each day's production, three (3) assemblies shall be randomly selected from the last ten (10) assemblies welded, and the weld pulled to destruction. If a reject occurs, that quantity of assemblies welded since the last acceptable check shall be rejected.

4.4.3.2.2 Lead to anode and cathode screen assembly. At the start of each day and after welding parameters have been set, weld three (3) anode and cathode lead to screen assemblies. Pull lead to destruction to determine weld integrity (para. 3.7.2). Do not proceed with production welding until all three (3) units are accepted. At the end of each four (4) hour period during the continuous production run and at the end of each day's production, three (3) units shall be randomly selected from the last ten (10) assemblies welded and the weld pulled to destruction. If a reject occurs, that quantity of assemblies welded since the last acceptable check shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.3.3 Radiograpic examination. Radiographic examination shall be performed on a sample of 72 cells from each lot for information only.

4.4.3.4 Electrolyte volume. At the start of each day, ten (10) electrolyte samples shall be taken and the volume of electrolyte checked. Do not proceed with production until all ten samples are accepted. At the end of each two (2) hour period during the continuous production run and at the end of each day's production, five (5) units shall be randomly filled and the volume of electrolyte checked. If any sample fails to contain the proper amount of electrolyte, that quantity of ampul assemblies filled since the last acceptable check shall be rejected. This check shall be made in accordance with para. 4.4.6.4.

4.4.3.5 Heat soak. Each ampul (9328782) shall be subjected to the test. Any ampul exhibiting cracks or leaking after the heat soak shall be rejected and removed from the lot. The test shall be performed as specified in para. 4.4.6.6.

4.4.3.6 Hardness. A sample of three (3) cases (9275592) shall be selected from each lot and tested for hardness. Failure of any one (1) sample to meet the requirement of 3.15 shall result in rejection of the lot. This test shall be performed as specified in 4.4.6.10.

4.4.4 Dimensional control of stamped/molded parts. In place of the normal sampling associated with the Classification of Defects, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a sample of five (5) parts shall be selected during the first hour of each day's production and during the last hour of each day's production (up to and including 8 hours) for dimensional inspection as a control of the tool during production. As a control of each tool during production, the above quantity of parts shall be inspected for at least the defects listed in their respective paragraphs of 4.4.2.1 thru 4.4.2.22. If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be inspected for each separate type of defect in accordance with the sampling plan and AQLs as specified in the referenced paragraphs. All tools removed from production because of some fault, may after reworking, be returned to production, providing they pass the qualification test above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.5 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.4.6). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6 herein.

4.4.6 Test methods and procedures.

4.4.6.1 Non-activation voltage. Batteries specified in 4.3.4 and 4.4.3.1 shall be measured for non-activation voltage using an acceptable digital millivoltmeter having a minimum impedance of 10^7. Any battery which measures more than the maximum requirement in paragraph 3.3 shall be classed defective and removed from further tests. Acceptable batteries may be used for the rise time and capacity test.

4.4.6.2 Rise time and capacity test.

4.4.6.2.1 First article test. Batteries specified in 4.3.4 shall be assembled in pairs in the molding cylinder (dwg. 9275635) to form a power supply assembly (dwg. 9275636). The assembly shall be loaded with the stab primer (dwg. 9296892), in accordance with procedures dwg. 9275634. The assemblies shall be divided into three (3) equal groups. One group shall be placed in a temperature chamber and conditioned at 145°F for a minimum of two (2) hours; one group shall be conditioned at room ambient for a minimum of two (2) hours; and one group shall be conditioned at -45°F for a minimum of two (2) hours. All temperatures shall be maintained within $\pm$ 5°F.

Each power supply assembly shall then be connected to a 13.2 $\pm$ 3% resistive load and activated. Activation shall be accomplished by applying a two (2) inch ounce force to the primer through the firing-pin (dwg. 9275628). Rise time shall be measured by an oscilloscope or other approved measuring equipment, and the voltage shall be measured and recorded at the time of activation and at one hour intervals during the duration of test to end of life (5 volts). Failure to comply with rise time and/or life requirements shall cause the power source assembly to be classified defective (destructive test).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.6.2.2 Lot acceptance test. Batteries specified in paragraph 4.4.3.1 shall be divided into three (3) equal groups. One group shall be placed in a temperature chamber and conditioned at +145°F for a minimum of one (1) hour; one group shall be conditioned at room ambient for a minimum of one (1) hour; and one group shall be conditioned at -45°F for a minimum of one (1) hour. (All temperatures ± 5°F).

Each battery, after conditioning, shall be connected to a 6.6K ± 3% resistive load and activated. Activation shall be accomplished by striking the center of the non-terminal end of the battery with a right circular steel cylinder having a mass of 0.2 ± .02 grams, and traveling at a velocity of 175 ± 30 feet per second. The diameter of the cylinder shall be 0.071 - .004 inch with corner breaks of .020 ± .005 inch. Rise time shall be measured by approved measuring equipment, and the voltage shall be measured and recorded at the time of activation and at one (1) hour intervals during the duration of the test. (Activation time shall be measured from the time the case is depressed.)

4.4.6.3 Radiographic examination. Inspection equipment and image quality of the radiographics for the examination shall be satisfactory to the procurement contracting officer. The direction of the x-ray beam employed in the inspection shall be perpendicular to the cylindrical wall of the battery, with the anode lead to screen welds positioned to the right. Retain x-ray films in the file during production. (Non-destructive test.)

4.4.6.4 Electrolyte volume. Individually, pre-weigh and record the weight of ten (10) ampul blanks. After filling, re-weigh the filled ampuls and calculate the weight of the electrolyte. Using the specific gravity of the electrolyte batch, calculate the volume. Repeat at intervals specified in para. 4.4.3.4 with five (5) blanks.

4.4.6.5 Leak test. Each battery shall be subjected to the leak test in accordance with ASTM-E 493-73 and shall meet the leak rate requirements specified on drawing 9328787. If the fine leak test equipment has the capability of gross leak detection, each battery shall be subjected to a leak test using this equipment. However, if this equipment is not capable of gross leak detection, each battery shall be subjected to both a fine leak test and a gross leak test.

The gross leak test shall be performed in accordance with MIL-STD-202, Method 112, Condition B. Any evidence of bubbles shall be cause for rejecting the leaking battery.

The fine leak shall be performed in accordance with ASTM-E 493-73, Method A. The volume V for calculating the actual leak shall be 0.6 cc. Any battery failing to meet the actual leak rate requirements of the drawing shall be rejected. (Non-Destructive Test)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.6.6 Heat soak. Each ampul assembly (9328782) shall be placed in a chamber heated to 165°F ± 5°F for 24 hours minimum. After the 24 hours has elapsed, the ampul shall be removed from the chamber and visually examined for cracks and leakage. Any unit exhibiting cracks or leakage shall be classed defective and removed from the lot. (Non-destructive test.)

4.4.6.7 Environmental testing. Batteries specified in 4.3.4 shall be assembled in pairs as specified in 4.4.6.2.1.

Batteries specified in 4.4.2.1 shall be assembled in pairs as specified in 4.4.6.2.1. The assembly shall be loaded without the stab primer. These batteries will only be subjected to the shock test. The batteries shall be checked for non-activation voltage in accordance with 4.4.6.1 after waiting a period of twenty-four (24) hours.

The following procedures shall be followed in performing the tests:

a. Shock - 7 ft. Drop - MIL-STD-331, Method 111 - The batteries shall be placed in pairs within a block of hardwood or other suitable material weighed to five (5) pounds ± 1/4 lb. The shape of the block shall be approximately a right circular cylinder two (2) inches high by five (5) inches in diameter. Three (3) assemblies shall be conditioned at -45°F for two (2) hours minimum and three (3) assemblies at ±145°F for two hours minimum. One (1) unit from each temperature group shall be dropped on the flat of the cylinder, one (1) unit from each temperature group shall be dropped on the outside of the cylinder approximately perpendicular to the longitudinal axis of the battery assembly, and one (1) unit from each temperature group shall be dropped on the outside of the cylinder approximately perpendicular to the ends of the battery.

b. Transportation vibration - MIL-STD-331, Test 123, except that the test shall be performed at ambient temperature.

After each environment, the batteries shall be checked for non-activation voltage in accordance with 4.4.6.1 after waiting a period of 24 hours. Units passing the non-activation voltage test shall then be tested for rise time and capacity in accordance with 4.4.6.2.1.

4.4.6.8 Ballistic rail gun test. Batteries specified in 4.3.4 shall be assembled in pairs in the M718/M741 electronic lens assembly (dwg. 9287150/9287151). The assembly shall be loaded without the stab primer. These batteries to be tested for acceleration shall be shipped to ARDEC where they shall be mounted in fixtures, representative of the mines, loaded into the appropriate projectile, and subjected to the zone environment of the ballistic rail gun. One-third of the quantity of batteries shall be tested at -25°F, one-third tested at +145°F, and one-third tested at room ambient (All temperature ± 5°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

After being subjected to the ballistic rail gun, the batteries shall be checked for non-activation voltage in accordance with 4.4.6.1.

4.4.6.9 Ballistic functioning. The number of batteries specified in 4.3.4 shall be shipped to a destination as designated by the contracting officer. These batteries shall be assembled in pairs into Non-Haz M741 mines, drawing 9317608. The test mines shall be loaded into projectiles in accordance with drawing 9317598 and MIL-P-63430. The twelve (12) projectiles shall be equally divided into three (3) groups and preconditioned at hot (+145°F), cold (-25°F), and ambient. The loaded projectiles shall be tested as described in MIL-P-63238 for non-hazardous mines except only self-destruct will be tested and all projectiles will be fired with the appropriate Zone 7 charge for the M109A1. Acceptance will be based on self-destruct functioning. If 9 (nine) or more mines out of the battery sample prematurely functions (SD -20%), the test sample shall be rejected. Inclusive in this accept-reject criteria are the duds due to battery failure. Duds due to S&A malfunctioning shall not be scored as a failure.

4.4.6.10 Hardness. Hardness of the cases shall be measured, using a Knoop hardness tester, by taking readings on the side and closed end. A minimum of three readings shall be made along the side, and three on the closed end, approximately 120° apart. Parts should be mounted and sectioned before taking 200 gm load Knoop impressions.

4.4.6.11 Cathode weight. A sample of cathodes (12556494) shall be selected from each lot in accordance with MIL-STD-105, Level II. Units failing to comply with the weight specified on the drawing shall be classed defective. (Non-destructive test.)

4.4.6.12 Electrical short test anode to cathode. After assembly each battery shall be tested for evidence of anode to cathode short using standard measuring equipment. (Short is defined as resistance of less than one (1) megohm.)

4.4.7 Surveillance units. The batteries designated for surveillance shall be stored at the contractors facility until contract completion and disposition made. The units shall be identified in a manner to provide traceability to month and year of manufacture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

5 PACKAGING

5.1 Preservation.

5.1.1 Level B. Batteries shall be packed for shipment in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level B. Batteries shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. Marking of the carton shall comply with MIL-STD-1169 and lot number with MIL-STD-1168.

5.4 Shipping. When parts from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. These batteries are designed for use as the reserve power supply for electronic mines.

6.2 Ordering data. See MIL-A-48078.

6.3 First article. Provisions for first article as specified in Sections 3 and 4.

6.4 Submission of ballistic first article sample. The ballistic first article sample shall be submitted to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, New Jersey 07806-5000.

6.5 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, New Jersey 07806-5000.

6.6 Subject term (key word) listing.

High Energy Density (HED) Battery
Ballistic Rail Gun
Battery Voltage Rise Time
Battery, Non-activation Voltage
Anode and Cathode
Battery, Ampul
Battery, Electrolyte

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A484)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
U S ARMY ARDEC
ATTN: AMSMC-QA
PICATINNY ARSENAL, NJ 07806-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-63524A	BATTERY, HED, SINGLE CELL, PRIMARY RESERVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202640-26104:27Z
Check the source to verify that this is the current version before use.